

Statistical Models in Computational Biology

Niko Beerenwinkel
Auguste Rimaite
Alexander Jung
Yueheng Zhang
Rudolf Schill

Due date: 25th March 2026 before 14:00

Problem 12: Intuition on time-reversible Markov chains (1 point)

Recall the concept of time-reversibility as defined in the lecture. Intuitively, a Markov chain is called time-reversible if it is in some sense indistinguishable from the same process viewed backwards in time, which you will prove in this exercise.

Let $\{X(t), 0 \leq t\}$ be a homogeneous continuous-time Markov chain on the finite state space $[K] = \{1, \dots, K\}$ with transition probabilities

$$T_{kl}(t) = \Pr(X(s+t) = l \mid X(s) = k), \quad k, l \in [K], t \geq 0.$$

Define its time-reversed process on a finite time-interval $[0, \tau]$ by

$$\{\hat{X}(t) := X(\tau - t), 0 \leq t \leq \tau\}$$

with transition probabilities

$$\hat{T}_{kl}(s, t) := \Pr(\hat{X}(s+t) = l \mid \hat{X}(s) = k),$$

for $0 \leq s \leq s+t \leq \tau$.

1. Assume that the original Markov chain has a stationary distribution π with $\pi_k > 0$ for all k and starts out stationary as $X(0) \sim \pi$. Show that the time-reversed process has transition probabilities

$$\hat{T}_{kl}(s, t) = \frac{\pi_l T_{lk}(t)}{\pi_k}$$

which do not depend on s . This means that the time-reversed process is homogeneous too and we may then write $\hat{T}_{kl}(t) := \hat{T}_{kl}(s, t)$.

2. Assume additionally that the original Markov chain is time-reversible:

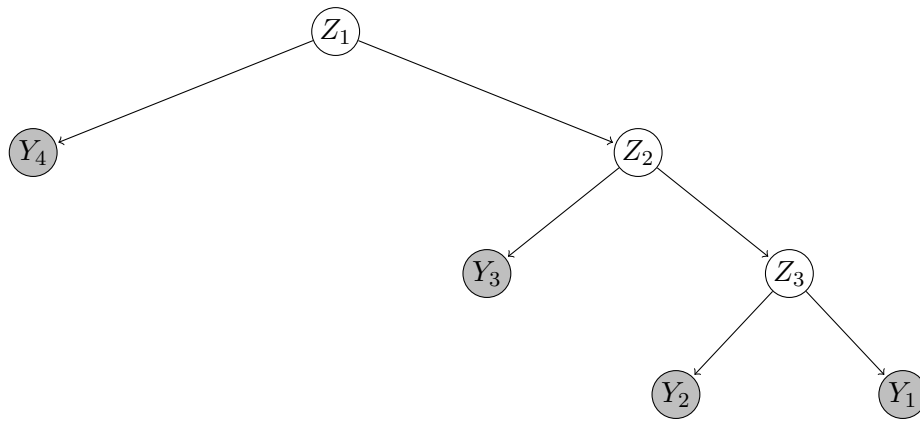
$$\pi_k T_{kl}(t) = \pi_l T_{lk}(t) \quad \text{for all } k, l \in [K], t \geq 0.$$

Show that

$$T_{kl}(t) = \hat{T}_{kl}(t) \quad \text{for all } k, l \in [K], 0 \leq t \leq \tau.$$

Problem 13: Phylogenetic trees as Bayesian networks

(3 points)



Consider the above phylogenetic tree T with observed variables $Y = (Y_1, \dots, Y_4)$ and hidden variables $Z = (Z_1, \dots, Z_3)$. Each variable Y_i or Z_i takes values in $\{A, C, T, G\}$. Assume that a prior $P(Z_1)$ on Z_1 is given.

1. Factorize the joint probability $P(Y, Z|T)$ of Y and Z conditioned on the tree topology T . Then provide the naive expression for the marginal probability $P(Y|T)$ when marginalizing over the hidden variables Z in a brute-force manner. How many summation terms¹ are required naively? (1 point)
2. Rearrange the expression $P(Y|T)$ according to Felsenstein's peeling algorithm such that the number of operations is minimized. How many summation terms are required at most in this form? (2 points)

Problem 14(data analysis): Learning phylogenetic trees from sequence alignment data (6 points)

The aim of this problem is to solve a forensic case by studying phylogenetic trees. For this purpose you will estimate the topology and the parameters of a phylogenetic tree from sequence alignment data using the R packages `phangorn` and `ape`. Learning a phylogenetic tree with these packages comprises the following steps:

- i. Calculate pairwise distances between the sequences in the alignment.
- ii. Generate a first tree topology based on neighbour joining.
- iii. Optimise the likelihood of a phylogenetic model with respect to branch lengths, nucleotide substitution rates, and tree topology. Phangorn uses heuristics to locally search the space of possible tree topologies and to find a topology with (locally) highest score.

We will look at a dataset collected in a health care institution in the suburbs of Paris. Each data point is the DNA sequence of a certain genomic region ('RT') of the HIV-1 virus extracted from HIV-1 positive persons. This dataset was collected in the framework of a virological investigation² requested by the authorities in order to determine whether a nurse-to-patient infection had occurred in the health care institution mentioned above. A patient was infected with HIV-1 during her stay at the institution. After screening the members of the staff who had had contact to the patient, the authorities found out that one male nurse and one female nurse were already HIV-1 positive during the time the patient was at the institution. Your task is to establish which nurse is more likely

¹here we are counting summation terms, not summations. E.g. $a + b + c$ counts as three summation terms.

²Goujon C. et al. *Phylogenetic analyses indicate an atypical nurse-to-patient transmission of Human Immunodeficiency Virus Type 1*. Journal of virology, 2000

to have infected the patient. Furthermore, you have to assess the likelihood that the patient was actually infected at the hospital based on the phylogenetic comparison to control HIV-1 strains from the Paris area. In order to assess the uncertainty associated with phylogenetic tree inference, you will compare the outcomes of the phylogenetic tree learning procedure across bootstrap resamples from the alignment.

1. Install and load the R packages `phangorn` and `ape`. Load the alignment `ParisRT.txt` into memory using the function `read.dna()`.
2. Create an initial tree topology for the alignment, using neighbour joining with the function `NJ()`. Base this on pairwise distances between sequences under the a. Kimura (1980), b. (Tamura and Nei 1993) and c. (Jukes and Cantor 1969) nucleotide substitution models, computed using the function `dist.dna()`. Plot the initial trees. (1 point)
3. Use the function `pml()` to fit the Kimura model (`model = "K80"`) to the above tree (a) and the alignment. Note that the function expects `data = phyDat(alignment)`. What is the log likelihood of the fitted model? (1 point)
4. The function `optim.pml()` can be used to optimise parameters of a phylogenetic model. Find the optimal parameters of the Kimura (1980) nucleotide substitution model whilst the other parameters are held *fixed*. What are the values in the optimised rate matrix? (1 point)
5. Optimise the Kimura model with respect to branch lengths, nucleotide substitution rates, and tree topology simultaneously. What is the log likelihood of the optimised model? (1 point)
6. The function `bootstrap.pml()` fits phylogenetic models to bootstrap resamples of the data. Run it on the optimised model from step 5, but pass the argument `optNni = TRUE` to allow for a different topology for each bootstrap run. What, exactly, is being resampled? (1 point)
7. Use `plotBS()` with `type = "phylogram"` to plot the optimised tree (from step 5) with the bootstrap support on the edges. Which nurse ("Mme_S" or "Mr_D") is more likely to have infected the patient "Mme_L"? (1 point)